

Research Proposal

Is Visual Grounding Decay Real? Separating Disengagement from Redundancy in Multimodal Chain-of-Thought

Working title (submission form): *Marginal Is Not Total: A Mediation Decomposition of Visual Dependence in Multimodal Reasoning*

Project codename: PRISM — Per-step Reliance, Indirect effects, Sufficiency, Mediation

1. Executive summary

The multimodal reasoning literature has converged on a stylised fact: as a vision-language model generates a longer chain of thought, its dependence on the image declines. This is called visual grounding decay, and roughly twenty papers between 2025 and September 2026 propose rewards, memory modules, or re-injection mechanisms to fix it.

Every measurement supporting that fact — attention mass, image-text mutual information, per-token dependence measures, blank-image ablation — is computed on a reasoning prefix that was itself generated **with the image present**. Visual information has therefore already entered the text. When you ablate the image at step 12 and the log-probability of step 12 barely moves, there are two explanations, and the field has only ever reported one:

1. **Disengagement.** The model has stopped using visual evidence.
2. **Redundancy.** The model already extracted what it needed and wrote it down; the image is now informationally redundant with its own prefix.

These have opposite engineering implications. Under (1) you should force the model to keep looking. Under (2) forcing it to keep looking is wasted compute at best and a reward-hackable objective at worst — which would explain why the empirical literature on grounding rewards is so inconsistent.

This project treats per-step visual dependence as a **causal mediation problem**, where the reasoning prefix is the mediator. We decompose total visual dependence into a controlled direct effect and an indirect effect carried by the prefix, measure both across models, benchmarks and training regimes, and determine what fraction of reported grounding decay is redundancy rather than disengagement. We then extend the decomposition into a per-step **necessity × sufficiency** taxonomy that distinguishes genuinely grounded steps from redundantly grounded, prior-driven, and confabulated ones.

The core contribution is diagnostic and is robust to outcome: whichever way the decomposition falls, the result corrects or hardens the field's central empirical claim. A calibration objective derived from the measurement is deferred to a second paper.

2. Rationale

2.1 Why this, and why now

Three things make this the right project at this moment.

The measurement space that looks open is not, and the space that looks closed is not. A systematic audit (September 2026) found that counterfactual and ablation-based reliance measurement is heavily occupied — VSI (2609.00868), vCT/vCCT (2609.06704), the blank-image arbiter (2606.31257), SGR (2603.06828), DeFacto (2509.20912, ICML 2026), CFPO (2606.23206, ICML 2026), DVRP (2601.06801). Proposing "a counterfactual reliance measure" in late 2026 is proposing the eleventh one. But the *confound shared by all of them* is unclaimed. That asymmetry is the opportunity.

The confound is load-bearing, not cosmetic. It does not merely add noise; it biases every existing measurement in the direction of the field's headline finding. Marginal dependence must fall as the prefix accumulates visual content, by construction, even for a model that never stops looking. A literature that has only measured the marginal quantity has been reading a partly mechanical artifact as a behavioural pathology.

The nearest neighbour stops one step short. "Who Drives the Probability Game of VLMs?" (2609.02000, Beijing Normal / Kuaishou, 2 September 2026) decomposes step-indexed Visual, Question and Prefix Causal Drive via backdoor adjustment and reports a Prefix–Visual imbalance score. It is the closest published work and it establishes that the prefix is a first-class causal driver. It does not make the marginal-versus-total distinction, does not contrast ablate-at-step-s against ablate-throughout, and does not reinterpret the decay literature. That gap is the paper. It is also a warning: one increment separates that group from this contribution, which sets the timeline.

2.2 Why the diagnostic, not the reward

The original line of work aimed at a step-level attention-aware process reward in GRPO. That space contains at least ten entries (Reflection-V, RAL, AVAR, VGPO, Faithful-MR1, LASER, SAYO, CFPO, DVRP, PAPO, Perception-R1), several accepted at ICLR 2026 and ICML 2026. Entering it now with a bidirectional variant is a 15%-acceptance proposition against well-resourced groups shipping at roughly 1.5 papers per week.

More importantly, a reward built on a confounded measure inherits the confound. If a substantial share of decay is redundancy, then every reward that penalises falling marginal dependence is penalising efficient reasoning. The measurement must come first on scientific grounds, not merely tactical ones.

2.3 Why it cannot produce a null result

- If total dependence declines as steps advance, grounding decay is real and this work supplies the first confounder-controlled evidence for it, hardening a claim the field currently holds on weak measurement.
- If total dependence is flat while marginal dependence falls, a large body of published diagnoses is partly artifactual, and roughly ten reward methods are optimising the wrong quantity.
- If the split is regime-dependent — real on sustained-visual-reference tasks, artifactual on symbolic ones — that is the principled version of the task-conditional finding in *Don't Blink* (2604.04207), which explicitly flags its own attention measure as correlational rather than causal and names causal intervention as future work.

All three are publishable. There is no configuration of results that wastes the compute.

3. Problem statement

Let a vision-language model π generate a reasoning trajectory $y = (y_1, \dots, y_S)$ conditioned on an image I and question q . The literature reports that a per-step measure of visual dependence declines in s , and interprets this as the model progressively ignoring the image.

Existing measures — attention mass over image tokens, the Visual Prompt Dependency Measure, blank-image log-probability differences, VSI-style perturbation sensitivity — share a common form: hold the prefix $y_{<s}$ fixed at the trajectory the model actually produced with the image, ablate the image, and measure the change at step s .

Holding $y_{<s}$ fixed is exactly the operation that defines a controlled direct effect in causal mediation. The prefix is a mediator through which the image influences step s . Conditioning on it blocks the indirect path. What the field measures is therefore the direct effect only, and its decline is fully consistent with a constant total effect and a growing indirect one.

Formally, the field has estimated CDE_s and reported it as if it were TE_s .

Three consequences follow, and this proposal addresses each:

1. **No published estimate of total per-step visual dependence exists.** We do not know whether grounding decay survives control for prefix redundancy.
2. **The necessity/sufficiency distinction is absent for visual evidence.** All existing measures are necessity measures (remove the image, does the output change). None asks whether the image alone suffices to produce a step. The causal necessity/sufficiency formalism has been applied to *textual* chain-of-thought (2506.09853, NeurIPS 2025) but not to visual evidence, so redundantly

grounded steps are indistinguishable from prior-driven ones under current instruments.

3. **Demand is estimated, never constructed.** Frameworks that ask whether a step *should* depend on vision estimate that quantity from a model with the same pathology under study (BICR's frozen probe, VEPA's frozen blind reader). No dataset supplies per-step visual demand as ground truth by construction.

4. Literature review

Scope: January 2024 – September 2026, emphasis on 2025–2026. Acceptance status distinguished throughout, since preprint prior art and proceedings prior art carry different weight under CVPR and ICCV policy.

4.1 The grounding-decay finding and its instruments

The claim rests on four measurement families, all correlational or all marginal.

Attention-based. Depth-wise attention decay to image tokens after early decoding layers (FastV and successors), visual attention sinks, and declining image–text mutual information (Favero et al., 2024) are the original evidence. *Seeing but Not Believing* (2510.17771) complicates this directly: deep layers reliably attend to the correct evidence regions while the model still answers incorrectly — models that perceive but do not believe. The attention-is-not-explanation objection therefore applies with full force in the multimodal setting, and any grounding measure resting on attention mass is answering the wrong question.

Per-token dependence. The Visual Prompt Dependency Measure, a Hellinger-distance formulation derived from Favero et al. and used in recent diffusion-reasoning work (2604.05497), shows the canonical early-high / late-low profile. It is a *marginal* measure and inherits the confound in full.

Per-sample counterfactual. VSI (2609.00868, 1 September 2026) blurs the question-relevant region and measures the shift in the next-token distribution, reporting that the assumption of visual use fails on 40–97% of samples across six VLMs and three benchmarks, with VSI ranks correlating across architecturally dissimilar models. The blank-image arbiter (2606.31257) uses a grey-blank control across fourteen models spanning 2B–27B and separates grounded, prior and inverted spatial regimes. Both are per-sample and therefore cannot speak to within-trajectory dynamics at all.

Per-step. SGR (2603.06828) is step-level but behavioural and video/agentic rather than a log-probability intervention. It is the closest existing per-step instrument and is not a mediation decomposition.

Direct diagnostic. *Don't Blink* (2604.04207) reports pervasive evidence collapse across MathVista, HallusionBench and MMMU-Pro on 2–8B reasoning VLMs, and finds that

low-entropy visually disengaged predictions are hazardous on sustained-visual-reference tasks but benign on symbolic ones. Its limitations section concedes the measure is correlational and names activation patching as the required next step. This proposal is a stronger version of that next step.

4.2 The reward and intervention literature this displaces

Ten-plus methods optimise some form of visual grounding signal inside RL: Reflection-V (2509.12132), RAL (2602.04884), AVAR (2603.03825, ICLR 2026), VGPO (2604.09349), Faithful-MR1 (2605.22072), LASER (2607.01707), SAYO (2602.08241), CFPO (2606.23206, ICML 2026), DVRP / *Thinking with Deltas* (2601.06801), PAPO (2507.06448), Perception-R1 (2506.07218, ICLR 2026), DeFacto (2509.20912, ICML 2026). Uniformly, they push grounding *upward*. Two papers document that this is not obviously correct — *Don't Blink* on task-conditional benignness, and *When Does a Video-Language Model Stop Watching?* (2606.22043) on dose-response and hysteresis in the grounding penalty — but neither builds a two-sided objective. Notably, DVRP maximises reasoning divergence from masked inputs while minimising divergence from perturbed inputs, and reports that policies maintain or even improve performance under blank images. That is precisely the observation that motivates asking whether the image was ever marginally necessary.

Multimodal Reward Hacking in Reinforcement Learning (2607.09492) supplies the missing warning: in a controlled 2B–32B sandbox across GRPO, RLOO and DAPO, outcome-only rewards reach a 48.1% reward-hacking rate, and even a 32B model retains a 54.9% worse rate under outcome-only rewards. No paper cross-evaluates the published grounding rewards under a shared independent causal measure.

Correction to prior audits. "PVM (Huang et al.)" is Persistent Visual Memory (2605.00814, Shanghai AI Lab / SJTU / CUHK), a learnable FFN-parallel retrieval module addressing visual signal dilution — not an RL reward method, and not blocking prior art for reward design. "Adversarial Reward Auditing" (2602.01750) is text RLHF reward hacking and is off-theme.

4.3 Inference-time control

ReGround (2608.04385) triggers image re-injection from an attention-entropy signature plus learned self-diagnosis. SIEVE (2603.14117) learns to self-revisit via RL. LookWise (2603.00171) uses token-level confidence as a sufficiency indicator. GazeVLM (2605.07817, IBM/ETH) emits gaze tokens controlling the causal attention mask. AIM-CoT (2509.25699, ACL 2026), Look-Back (2507.03019, AAAI 2026), Qwen Look Again (2505.23558), CoRGI (2508.00378) and VLM-R³ (2505.16192) round out the family.

Every trigger in this family is attention-based, confidence-based, or learned self-diagnosis. None uses a counterfactual reliance signal. This is retained as the method coda in §6, Phase 5.

4.4 Constructed-demand resources

Minimal-pair and controlled-perturbation datasets exist: VisMin (NeurIPS 2024; object, attribute, count and spatial minimal changes), CLEVR-HOPE (2311.08695), VisualFLIP (2606.07872; same-question image pairs with deterministic answer flips and a Collapse Rate), the VB Visibility Benchmark (2603.06680; 2×2 minimal image × text edits), CCTVBench (2604.20460). All provide per-*sample* answer flips. None provides per-*step* visual demand across a reasoning trajectory. Our constructed-demand set (§5.4) fills this and is the released artifact.

4.5 Positioning table

Nearest neighbour	What it has	What it lacks that we supply
2609.02000 (PCD/VCD)	Step-indexed prefix vs visual causal drive	Marginal/total distinction; no reinterpretation of decay
VSI (2609.00868)	Cross-model counterfactual reliance	Per-sample only; no within-trajectory dynamics
PDM (2604.05497)	Per-step marginal dependence	Confounded by prefix; no decomposition
SGR (2603.06828)	Per-step grounding rate	Behavioural, video; not a log-prob intervention
<i>Don't Blink</i> (2604.04207)	Task-conditional regimes	Correlational by its own admission
2506.09853 (NeurIPS 2025)	Necessity/sufficiency for text CoT	Not applied to visual evidence
VisualFLIP / VisMin	Minimal-pair constructed demand	Per-sample, not per-step

5. Method

5.1 Three interventions, one held-out step

For each step s of a trajectory generated normally under (I, q) , we score the same token sequence y_s under four teacher-forced conditions. Two axes are manipulated: image present or ablated, and prefix intact or visually scrubbed.

Condition	Image	Prefix	Log-likelihood
C_1	I	$y_{<s}$	$\ell_1 = \frac{1}{ y_s } \log \pi(y_s I, y_{<s})$
C_2	$\tilde{I}$	$y_{<s}$	ℓ_2
C_3	I	$\text{scrub}(y_{<s})$	ℓ_3
C_4	$\tilde{I}$	$\text{scrub}(y_{<s})$	ℓ_4

From these:

$$M_s = \ell_1 - \ell_2 \quad (\text{controlled direct effect — what the field measures})$$

$$T_s = \ell_1 - \ell_4 \quad (\text{total visual dependence})$$

$$C_s = T_s - M_s = \ell_2 - \ell_4 \quad (\text{prefix-carried / indirect visual dependence})$$

$$S_s = \ell_3 - \ell_4 \quad (\text{visual sufficiency — dependence when the prefix cannot help})$$

Redundancy Ratio: $R_s = C_s / \max(T_s, \epsilon) \in [0,1]$.

Every quantity is a length-normalised log-likelihood difference in nats. We deliberately do not squash through a sigmoid: an arbitrary squashing makes magnitudes incomparable across models and step lengths, which would defeat a cross-model protocol. Where a bounded quantity is needed we report the Jensen–Shannon divergence between next-token distributions as a secondary measure, following VSI.

Why this is FlashAttention-compatible. All four conditions are ordinary forward passes with different inputs. No attention extraction, no custom masks, no position-dependent image visibility. Ablation is implemented by substituting the image, and prefix manipulation by substituting text. This matters: retaining FlashAttention roughly thirds the memory footprint at Qwen-scale image token counts.

5.2 The headline test

Fit per-trajectory regressions of M_s and T_s on normalised step index $\hat{s} = s/S$, yielding slopes β_M and β_T . The literature's claim is $\beta_M < 0$. Our test:

- $\beta_M < 0$ and $\beta_T < 0$, similar magnitude $\rightarrow$ decay is real; existing measurement was adequate.

- $\beta_M < 0$ and $\beta_T \approx 0$, with R_s increasing in $s \rightarrow$ decay is substantially a redundancy artifact.
- Regime-dependent split $\rightarrow$ the causal version of the Don't Blink task-conditional result.

Report mixed-effects models with random intercepts by item and by model, bootstrap CIs over items, and effect sizes rather than bare p -values. Pre-register the three outcome interpretations before running the full sweep.

5.3 The scrub operator — the main methodological risk

scrub($\cdot$) must remove image-specific content from the prefix while preserving its logical and symbolic structure. This is where the paper can fail, so it is gated and audited rather than assumed. Four operators, all reported:

- Entity masking. Replace visual referents (objects, attributes, spatial relations, transcribed values) with typed placeholders — [OBJ], [VAL], [REL] — via a constrained extraction model. Most conservative; preserves syntax exactly.
- LLM rewrite. A text-only model rewrites the prefix preserving inferential structure and removing image-derived facts. Most natural; highest leakage risk.
- Blind regeneration. Regenerate a prefix of matched length from question only, under the same model with the image ablated. Cleanest causally; changes the text most.
- Value randomisation. Preserve all structure but replace image-derived numeric and categorical values with plausible alternatives. Isolates value-carrying from structure-carrying content.

Scrub-fidelity audit (main paper, not appendix). On 600 sampled prefixes, two independent annotators score each scrub on (a) visual content removed, (b) logical structure preserved, on 3-point scales; report Cohen's κ . Automatic complement: a text-only model attempts to recover the image's key attributes from the scrubbed prefix; recovery accuracy should approach chance.

Gate G1: mean "visual content removed" $\geq 2.5/3$ with $\kappa \geq 0.6$, and automatic recovery within 5 points of chance, for at least two operators. Failing this, the total-effect estimate is not trustworthy and the project falls back to the blind-regeneration operator alone with a widened uncertainty statement.

Cross-operator agreement (Kendall's τ over T_s rankings) is reported as a finding in its own right — disagreement between scrub operators would itself be informative and is not currently in the literature.

5.4 Image ablation operators

Five, all reported, none privileged:

- 1. Mean-pixel fill
- 2. Matched-statistic Gaussian noise
- 3. Patch shuffle
- 4. Unrelated real-image substitution
- 5. Region blur (question-relevant region only, VSI-style)

Operator 4 is *not* the default. Swapping in a different real image changes what the model confidently answers *about*, so its Δ conflates reliance with distractor interference. Operator 5 is the most surgical and is used for the headline numbers; 1–4 establish robustness. Cross-operator Kendall's τ reported.

5.5 The necessity × sufficiency taxonomy

Binarise T_s (necessity) and S_s (sufficiency) at per-model medians, with a sensitivity sweep over thresholds.

	Ss high (image alone suffices)	Ss low
T_s high	Grounded — genuinely vision-driven	Compositional — needs image and prior context jointly
T_s low	Redundantly grounded — image would suffice but prefix already carries it	Prior-driven — vision irrelevant

A third axis, visual assertion (does the step assert an image-specific fact, per annotation), separates benign prior-driven steps from confabulated ones: low T_s , low S_s , high visual assertion. Confabulation rate becomes a reportable hallucination metric with a causal rather than lexical definition.

This is the first application of the necessity/sufficiency formalism — established for textual CoT at NeurIPS 2025 — to visual evidence at step granularity.

5.6 Policy-independent demand estimate

d_s is computed by the same four-condition procedure under a frozen reference VLM from a different family than the model under study. Cross-family choice is deliberate: a same-family reference shares the pathology being measured, which is the flaw in using a policy's own reference model as a demand oracle.

Two validations, and the second is the one that matters:

- **Rank validity:** AUROC of d_s against human perceptual/symbolic/mixed labels. Gate G2: AUROC ≥ 0.80 .
- **Magnitude validity:** mean absolute error of d_s against constructed demand on the PRISM-Minimal set ($\backslash S5.7$), where per-step demand is known by design. Gate G3: calibration MAE within a pre-specified band.

This distinction is the fix for a real hole in the earlier version of this project. Rank validation licenses only rank-based objectives (a Spearman shape term). An absolute-difference objective such as $\mathbb{E}_s |\rho_s - d_s|$ requires calibrated magnitudes. If G3 fails, we report rank-based results only and say so plainly, rather than quietly using a rank-validated estimator in a magnitude-sensitive metric.

5.7 PRISM-Minimal: demand by construction

A $\sim 2,000$ -item released benchmark in which per-step visual demand is a design parameter rather than an estimate.

Construction

Each item is a $(q, I^{(0)}, \{I^{(k)}\})$ tuple where $I^{(k)}$ differ from $I^{(0)}$ in exactly one attribute (colour, count, spatial relation, transcribed value, object identity). Question templates are authored so that the reference solution has a known step structure, and for each step it is determinable by construction which single-attribute edits flip that step's correct content.

Ground-truth demand: $d_s^* =$ the fraction of the item's minimal-edit set that changes step s 's correct content. A step whose content is invariant to every edit has $d_s^* = 0$; a step determined by one attribute has demand concentrated on that edit.

Sources

Synthetic CLEVR-style scenes for full control ($\sim 1,200$ items) plus real-image minimal pairs adapted from VisMin and VisualFLIP (~ 800 items) so results are not a synthetic-domain artifact. All items carry step-aligned reference solutions.

Splits

400 dev / 1,600 test; a held-out 200-item probe set for training-time monitoring in Paper 2.

This is the NeurIPS Datasets & Benchmarks artifact and the answer to the strongest reviewer objection against every demand-estimation framework published to date.

6. Step-by-step methodology

Phase 0 — Audit refresh and infrastructure

- **Forward-citation sweep** on 2609.02000, 2609.00868, 2609.06704, 2606.23206, 2603.06828, 2604.04207. Monthly thereafter. The prefix-decomposition idea is one increment from the PCD/VCD group; this is the primary scoop vector.
- **Reproduce** the marginal decay curve (M_s falling in s) on Qwen2.5-VL-7B / MathVista as a sanity anchor against published PDM-style results.
- **Build** the four-condition scoring harness. Verify FlashAttention retention and measure per-step cost.
- Gate G0: marginal decay reproduces. If it does not, the premise of the entire subfield is in question and the paper changes shape (which would be a bigger result).

Phase 1 — Scrub operator development and audit Implement four scrub operators.

- Human audit on 600 prefixes, two annotators, κ reported. Automatic recovery probe.
- **Gate G1** (§5.3). This gate is early and hard by design: everything downstream depends on it.

Phase 2 — The decomposition study (Weeks 5–8)

Full sweep of M_s, T_s, C_s, S_s, R_s across:

- **Models (6)**: Qwen2.5-VL-7B, Qwen3-VL-8B, InternVL3-8B, Gemma-3-12B-IT (or LLaVA-OneVision-7B), one RL-tuned reasoning checkpoint (Visual-RFT or R1-VL class), one 2–3B model for the scale axis. Cross-family coverage is required for the protocol claim.
- **Benchmarks (7)**: MathVista, MathVerse, MMMU-Pro, MMStar (vision-indispensable), HallusionBench, Visual-CoT, ChartQA. Split by the sustained-visual-reference versus symbolic-transition taxonomy so results are directly comparable to Don't Blink.
- **Ablation operators (5) × scrub operators (4)** on a reduced grid for the sensitivity analysis; headline numbers on region-blur × entity-mask with all cells reported in appendix.

Primary outputs: β_M versus β_T by model and benchmark type; R_s trajectories; cross-operator τ .

Phase 3 — Taxonomy, annotation and demand validation (Weeks 7–10, overlapping)

- **Annotate** 1,500 steps: perceptual / symbolic / mixed, plus a visual-assertion flag. Two annotators on 100% (not a subset), κ reported per label. Adjudication protocol documented. No GPT-only labelling for the primary set — a VLM

annotator shares the priors under study and makes the validation circular; an LLM pre-pass is used only to propose labels for human confirmation.

- **Compute** the necessity $\times$ sufficiency quadrant distribution by benchmark type and step position. Confabulation rate.
- **Gate G2** (AUROC ≥ 0.80) on the frozen cross-family demand estimator.

Phase 4 — PRISM-Minimal construction and magnitude validation (Weeks 6–12, parallel track)

- Author templates and generate synthetic items; adapt real-image minimal pairs; produce step-aligned reference solutions.
- Human verification of a 300-item sample for edit minimality and step alignment.
- **Gate G3:** calibration MAE of $d_{s\$}$ against $d^{\star}_{s\$}$ within the pre-specified band.
- Release with datasheet, licence, and a leaderboard-ready evaluation script.

Phase 5 — Utility demonstration: counterfactual-triggered re-injection (Weeks 11–13)

A deliberately small method section proving the measurement is actionable, not a new method paper.

At inference, compute a cheap online approximation of $T_{s\$}$ (two forward passes on the current step only, region-blur operator). When $T_{s\$}$ falls materially below $d_{s\$}$ on a step flagged high-demand, trigger image re-injection and regenerate the step. Compare against ReGround's learned self-diagnosis trigger and Look-Back's implicit refocusing on matched compute.

This occupies the one unclaimed slot in the inference-time control family — every published trigger is attention-based, confidence-based, or learned self-diagnosis. If it works it is a clean secondary result; if it does not, it is reported as a negative result and the measurement paper is unaffected.

Hard rule: no new method ideas after Phase 5 begins.

Phase 6 — Writing, adversarial internal review, submission (Weeks 13–15)

Internal red-team pass specifically targeting: scrub leakage, the magnitude-versus-rank validity distinction, and whether the mediation framing survives a reviewer who knows causal inference properly. Recruit one reviewer outside the multimodal area for the second.

7. Datasets

Dataset	Role	Size used	Notes
MathVista	Symbolic-transition reasoning	1,000	Direct comparability to <i>Don't Blink</i> , DVRP
MathVerse	Symbolic, vision-variant subsets	800	Vision-dominant vs text-dominant splits are natural demand contrasts
MMMU-Pro	Mixed, hard	800	Vision-only setting available
MMStar	Vision-indispensable	1,500	Curated to defeat text-only shortcuts; the critical control
HallusionBench	Hallucination / confabulation axis	800	Anchors the visual-assertion flag
Visual-CoT	Sustained visual reference, step-annotated	1,000	Existing step structure and bounding boxes
ChartQA	Value-transcription heavy	600	Maximal expected prefix redundancy — the strongest test case
PRISM-Minimal	Constructed per-step demand	2,000	Released artifact
VisMin / VisualFLIP	Source pools for real-image minimal pairs	—	Licence-checked before adaptation

Total scored trajectories $\approx 8,500$, times four conditions, times the operator grid on the reduced sweep.

Compute budget. Four teacher-forced forward passes per step, per trajectory. At \$8 that is roughly 32 forward passes per item on the headline grid. This is the dominant cost and must be planned for — it is substantially larger than the attention-extraction cost that FlashAttention retention saves, and prior versions of this project understated it. No RL training is required for Phases 0–4, which is what makes the timeline feasible.

8. Evaluation metrics

8.1 Primary

Metric	Definition	What it establishes
β_M , β_T	Slopes of marginal and total dependence on normalised step index	The headline decomposition
R_s	C_s / T_s , redundancy ratio	Fraction of dependence carried by the prefix
$\Delta\beta = \beta_T - \beta_M$	Slope gap	Direct measure of how much reported decay is artifactual
Quadrant distribution	Fractions in grounded / redundant / compositional / prior-driven	The taxonomy result
Confabulation rate	Low T_s , low S_s , visual assertion true	Causal hallucination measure

8.2 Validation and integrity

Metric	Gate
Scrub fidelity (human, 2 annotators, κ)	G1: $\geq 2.5/3$, $\kappa \geq 0.6$
Automatic attribute recovery from scrubbed prefix	Within 5 points of chance
Cross-scrub-operator agreement (Kendall's τ)	Reported; disagreement is a finding
Cross-ablation-operator agreement (Kendall's τ)	Reported
Demand estimator rank validity (AUROC vs human labels)	G2: ≥ 0.80
Demand estimator magnitude validity (MAE vs $d^{\star}_s$)	G3: pre-specified band
Annotation agreement (κ , per label type)	Reported per label

8.3 Secondary and control

- **Blind-drop Δ** — $\text{accuracy}(\text{image}) - \text{accuracy}(\text{image removed})$. A well-grounded model should collapse when blinded. Retained as an independent sanity anchor.

- **Image-swap consistency.**
- **Step-count normalisation controls** — mean \$\$\$, tokens per step, reported to exclude the "emit more, shorter steps" confound.
- **JS divergence variant** of each dependence measure, as a bounded cross-check on the log-likelihood formulation.
- **Correctness interaction** — all quadrant statistics reported separately for correct and incorrect trajectories, since conditioning on correctness is where several published contradictions originate.

8.4 Phase 5 only

- Accuracy delta versus ReGround and Look-Back at matched inference compute.
- Trigger precision/recall against human-labelled genuine grounding failures.
- Overhead in wall-clock and forward passes.

9. Threat model

Failure mode	Defence	Canary
Scrub leaks visual content, deflating $T_{s\$}$	Four operators; human audit; automatic recovery probe	Recovery accuracy above chance
Scrub destroys logical structure, inflating $T_{s\$}$	Structure-preservation annotation score	Entity-mask vs LLM-rewrite disagreement
Ablation is OOD, so Δ measures novelty not reliance	Region-blur as headline operator; five-way comparison	Cross-operator τ collapse
Demand estimator inherits reference model's pathology	Cross-family reference; magnitude validation on constructed demand	G3 MAE
Rank-validated estimator used in magnitude-sensitive metric	G2/G3 separation; rank-only reporting if G3 fails	Explicit in results section
Effects are model- or benchmark-specific	Six models across four families; seven benchmarks split by taxonomy	Random-effects variance components

Failure mode	Defence	Canary
Synthetic-domain artifact in PRISM-Minimal	40% real-image items; results reported separately by source	Synthetic/real divergence

11. Risk register

Risk	Severity	Mitigation
Scrub operator fails audit (G1)	High	Fall back to blind regeneration alone; widen uncertainty; the taxonomy (§5.5) survives on M_s/S_s and remains publishable
PCD/VCD group publishes the marginal/total decomposition first	High	Monthly forward-citation refresh; early arXiv; the taxonomy and PRISM-Minimal are separable contributions that survive
$\beta_T \approx \beta_M$ — decay is entirely real	Medium	Publishable as the first confounder-controlled confirmation; reframe as hardening, not correcting
Demand magnitude validation fails (G3)	Medium	Report rank-only results; Paper 2's absolute-difference objective is dropped in favour of the Spearman shape term
Annotation quality insufficient	Medium	Two annotators at 100% coverage, documented adjudication, κ gates
Compute budget overrun on the four-condition grid	Medium	Reduce to 3B models for the full grid, 7–8B for headline; drop operator cells before dropping models
Reviewers want a method, not a diagnostic	Medium	Phase 5 inference-time result as the coda; NeurIPS D&B and TMLR are structurally more tolerant than CVPR

12. Contribution statement

1. **The first mediation decomposition of per-step visual dependence in multimodal chain-of-thought**, separating controlled direct, indirect and total effects, and the first evidence on whether reported grounding decay survives control for prefix redundancy.

2. **A per-step necessity × sufficiency taxonomy of visual evidence** — grounded, redundantly grounded, compositional, prior-driven — extending to vision a causal formalism previously applied only to textual reasoning, plus a causally defined confabulation rate.
3. **PRISM-Minimal**, a released benchmark supplying per-step visual demand as ground truth by construction rather than by model estimation, addressing the standing objection to every demand-estimation framework published to date.
4. **A validated, policy-independent, cross-family demand estimator**, with rank *and* magnitude validity reported separately and gated.
5. **A counterfactual-triggered inference-time re-injection controller**, occupying the one unclaimed slot in a family whose triggers are otherwise attention-based, confidence-based, or learned self-diagnosis.

Appendix A — Verified reference ledger

Acceptance status distinguished; unverified items flagged. All arXiv IDs confirmed against title, authors and date.

Diagnosis and measurement. VSI / Visual Insensitivity Gap (2609.00868, preprint) · Counterfactual Tests for CoT Faithfulness, vCT/vCCT (2609.06704, preprint) · Decodable Is Not Grounded (2606.31257, preprint) · Step-Level Visual Grounding Faithfulness, SGR (2603.06828, preprint) · Who Drives the Probability Game, PCD/VCD (2609.02000, preprint) · Don't Blink (2604.04207, preprint, UAI submission) · When Does a Video-LM Stop Watching (2606.22043, preprint, video modality) · Seeing but Not Believing (2510.17771) · LaViT (2601.10129, preprint; distillation method, not a diagnostic) · Reasoning Dynamics and the Limits of Monitoring Modality Reliance (2604.14888, COLM 2026) · Draft-and-Refine (2511.11005)

Reward and RL methods. CFPO (2606.23206, **ICML 2026**) · DeFacto (2509.20912, **ICML 2026**) · AVAR (2603.03825, **ICLR 2026**) · Perception-R1 (2506.07218, **ICLR 2026**) · Reflection-V / Look Again Think Slowly (2509.12132) · RAL (2602.04884) · VGPO (2604.09349) · Faithful-MR1 (2605.22072) · LASER (2607.01707) · SAYO / Do MLLMs Really See It (2602.08241) · DVRP / Thinking with Deltas (2601.06801) · PAPO (2507.06448) · Visual-RFT (2503.01785, ICCV 2025) · VLM-R1 (2504.07615) · VEPA (2606.17678) · BICR (2605.10893)

Inference-time control. ReGround (2608.04385, venue unconfirmed) · AIM-CoT (2509.25699, **ACL 2026 Main**) · Look-Back (2507.03019, **AAAI 2026**) · SIEVE (2603.14117) · LookWise (2603.00171) · GazeVLM (2605.07817) · Qwen Look Again (2505.23558) · CoRGI (2508.00378) · VLM-R³ (2505.16192)

Reward hacking. Multimodal Reward Hacking in RL (2607.09492) · Reward Hacking in the Era of Large Models (2604.13602, survey) · Disentangling Perception and Reasoning via Reward Design (2601.00215)

Causal formalism and benchmarks. Causal Sufficiency and Necessity Improves CoT Reasoning (2506.09853, **NeurIPS 2025**) · VisMin (**NeurIPS 2024**) · VisualFLIP (2606.07872) · VB Visibility Benchmark (2603.06680) · CLEVR-HOPE (2311.08695) · CCTVBench (2604.20460)

Corrections to earlier audits. Persistent Visual Memory (2605.00814) is a learnable memory module, **not** an RL reward method, and was previously miscited as "PVM (Huang et al.)" in the attention-reward family · "Adversarial Reward Auditing" (2602.01750) is text RLHF reward hacking, off-theme · ReGround is self-diagnosis plus re-injection, **not** GRPO with an asymmetric reward · DVRP is an RL method with masked and perturbed visual triplets, **not** merely a diagnostic reporting blank-image gains.

Caveat. Most 2026 items are preprints. CVPR and ICCV policy does not require comparison against unpublished preprints, and anything posted after 15 September 2026 is contemporaneous for CVPR 2027. Occupied-on-arXiv and occupied-in-proceedings are different conditions and are treated differently throughout this proposal.